MIL-R-11073C(MU)
2 September 1966
SUPERSEDING
MIL-R-11073B(MU)
27 August 1963

MILITARY SPECIFICATION

ROTATING BAND BLANKS, SINTERED IRON

1. SCOPE

1.1 Scope.- This specification covers rotating band blanks made of sintered iron for use on all sizes and types of projectiles.

1.2 Classification.- Rotating band blanks shall be of the following classes, as specified:

Class	Density (g/cc)
1	5.30-5.59
2	5.60-5.89
3	5.90-6.19
4	6.20-6.49

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

STANDARDS

Federal

Fed. Test Method Std. No. 151 — Metals; Test Methods

Military

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

AMERICAN SOCIETY FOR TESTING MATERIALS

ASTM B 328-60 — Density and Interconnected Porosity of Sintered Powder Metal Structural Parts and Oil-Impregnated Bearings

FSC 1395

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(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia 3, Pa.)

3. REQUIREMENTS

3.1 Chemical composition.- The chemical composition of the band blanks, after sintering, shall be within the limits shown in Table I.

TABLE I - Chemical composition

Composition	Percent
Total iron, min.	97.75
Combined carbon, max.	0.15
Total silicon, max.	0.30
Total aluminum, max.	0.20
Sulfur, max.	0.05
Phosphorus, max.	0.05
Manganese, max.	0.60
Total of all other elements determined by difference, max.	0.90

3.1.1 The contractor shall furnish an analysis of each band blank lot showing the percentage of the constituents listed above.

3.2 Density.- The sintered iron rotating band blanks, before wax impregnation, shall meet the density requirements shown in Table II for the class specified.

3.3 Dimensions.- The rotating band blanks shall conform to the dimensions and tolerances specified on the applicable drawing.

3.4 Tensile strength.- The tensile strength of the band blanks shall meet the requirements shown in Table II for the class specified.

TABLE II - Property requirements

Class	Density (g/cc)	Tensile strength, psi, min.	Ductility - % min. expansion of band circumference
1	5.30-5.59	11000	3
2	5.60-5.89	13000	4
3	5.90-6.19	16000	5
4	6.20-6.49	20500	6.5

3.5 Ductility.- Wax impregnated rotating band blanks shall meet the ductility requirements shown in Table II for the class specified.

3.6 Secondary operations.- Secondary operations, such as sizing, will be permitted, however, the suppliers' bands blanks must still meet the maximum allowable density variation and the tensile and the ductility requirements.

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3.7 Wax impregnation.- The sintered blanks shall be impregnated with a micro-crystalline wax having a melting point range of 190° to 200° Fahrenheit (F). The wax content of the blank after impregnation shall meet the requirements shown in Table III for the class specified.

TABLE III - Wax content requirements

Class	Wax content % by Wt., min.
1	3.25
2	2.50
3	1.75
4	1.00

3.8 Workmanship.- The finished rotating band blanks shall be essentially uniform in density and free from chipped edges, cracks or other defects detrimental to serviceability. Superficial oxide films on the blanks shall not be cause for rejection, provided the degree of oxidation is not considered by the contracting officer to be detrimental to proper functioning of the band.

4. QUALITY ASSURANCE PROVISIONS

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to the prescribed requirements.

4.2 Lot.- A lot shall consist of rotating band blanks submitted for inspection at the same time, made from the same powder lot or blend to the same dimensions, and sintered at the same temperature in the same furnace. The use of a continuous furnace is permissible providing the temperature and atmosphere do not change through the lot. In addition, a lot shall not exceed 50,000 blanks.

4.3 Sampling.

4.3.1 Sampling for chemical analysis.- If, in the opinion of the contracting officer, a check chemical analysis of the band lot is necessary, the contractor shall submit a sample of the band blanks from the lot in question. Samples shall meet the chemical requirements in 3.1. The sample size shall be sufficient to conduct the required tests.

4.3.2 Sampling for mechanical property tests.

4.3.2.1 Samples for tension test.- At least ten samples shall be selected from each band lot of 10,000 pieces or less and at least twenty samples from each lot of over 10,000 pieces.

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4.3.2.2 Samples for ductility test.- At least twenty samples shall be selected from each band lot of 10,000 pieces or less and at least forty samples from each lot of over 10,000 pieces.

4.3.3 Samples for density test.- At least five samples shall be selected from each band lot of 10,000 pieces or less and at least ten samples from each lot of over 10,000 pieces.

4.3.4 Samples for wax content test.- At least five samples shall be selected from each band lot of 10,000 pieces or less and at least ten samples from each lot of over 10,000 pieces.

4.3.5 Samples for visual and dimensional examination.

4.3.5.1 Samples for visual examination.- Unless otherwise specified in the contract or order, samples for visual examination shall be obtained in accordance with Military Standard MIL-STD-105, Inspection Level II, AQL (acceptable quality level) = 4.0 percent defective.

4.3.5.2 Samples for dimensional examination.- Unless otherwise specified in the contract or order, samples for dimensional examination shall be obtained in accordance with Military Standard MIL-STD-105, Inspection Level II, AQL = 4.0 percent defective. The samples selected for dimensional examination may be the same as those selected for visual examination.

4.4 Examination.

4.4.1 Visual.- Each sample unit selected in accordance with 4.3.5.1 shall be examined to determine compliance with the requirements for workmanship (see 3.8).

4.4.2 Dimensional.- Each sample unit selected in accordance with 4.3.5.2 shall be measured to determine compliance with the dimensional requirements (see 3.3).

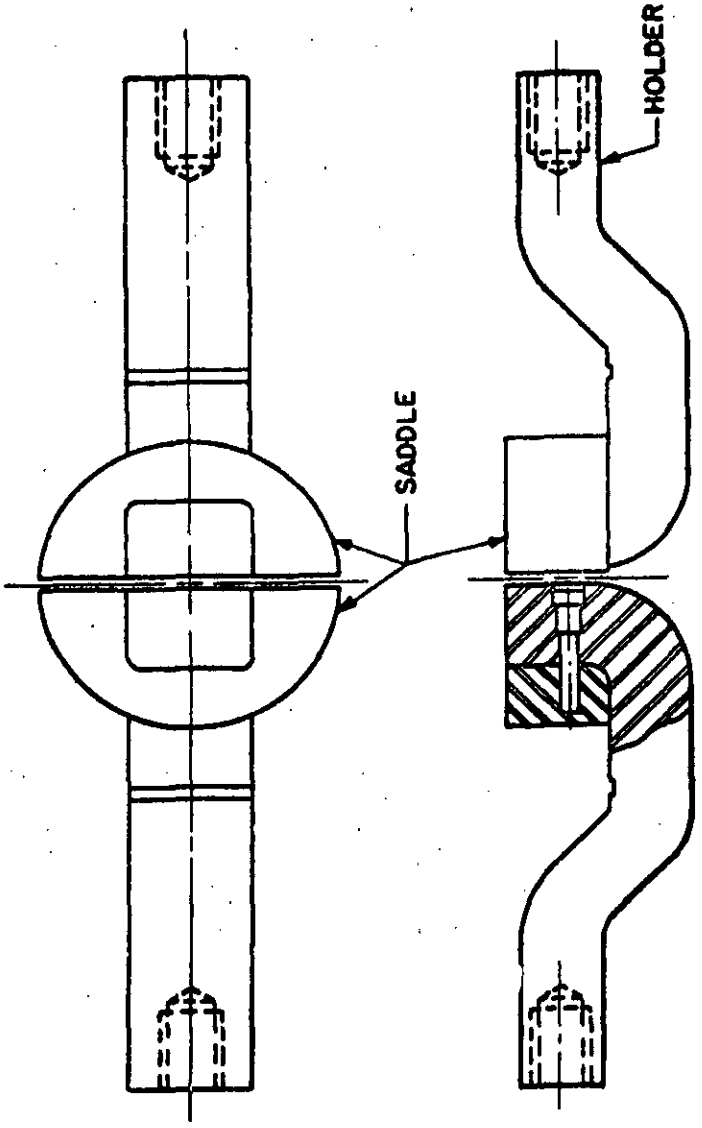
4.4.3 Preparation for shipment.- Examination of the preservation, packing and marking for shipment shall be made for conformance to the requirements of Section 5.

4.5 Test procedures.

4.5.1 Chemical analysis.- Unless otherwise specified in the contract or order, chemical composition shall be determined in accordance with method III or 112 of Fed. Test Method Std. No. 151.

4.5.2 Tension test.- The hand blank samples shall be tested in tension, on a "saddle fixture" as shown in Figure 1. This test should be performed at a speed of 0.250 inches per minute. The tensile strength is calculated by dividing the load at fracture by twice the cross-sec-

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Diameter of the saddle = Minimum diameter of the rotating band blank minus 0.020 inches minus 0.010 inches (machine tolerance)

TENSILE TEST FIXTURE ASSEMBLY FOR ROTATING BANDS

FIGURE 1

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tional area of the band blank. The tension test results for all specimens representing a lot shall comply with the requirements of Table II.

4.5.3 Ductility test.- Wax impregnated band blanks shall be placed on a tapered mandrel (see 4.5.3.1) and pushed downward by hand until the band blank is firmly wedged on the taper of the mandrel. The diameter of the mandrel in the plane of the leading face of the band blank shall be taken as the inside diameter of the band at the start of the test. The blank shall then be pushed along the mandrel, without fracture, a sufficient distance to determine compliance with the requirements of 3.5.

4.5.3.1 The mandrel shall have a diametric taper of 0.020 inches per inch. The mandrel shall be hardened and ground to a surface finish not greater than 16 micro-inches, RMS.

4.5.3.2 Any convenient method of forcing the blank at a uniform rate of from 2 to 4 inches per minute along the length of the mandrel may be used.

4.5.3.3 The inside diameter of the blank and the outside diameter of the mandrel should be lubricated with a suitable lubricant, preferably with a saturated solution of stearic acid in benzene.

4.5.4 Density test.- The density of a band shall be checked using the procedure described in *ASTM Test Method B328-60*. This shall include checking the density of small segments cut along the width of the band blank to determine compliance with the density limits of the class as specified in Table II. The band shall be divided into three segments along the width of the band blank. The density of the end segments of the band shall be uniform to within .2 gms/cc. However, the overall density of the band blank shall fall within the density limits as specified in Table II.

4.5.5 Wax content test.- The percent by weight of wax in blanks may be checked by any convenient method, such as weighting the blanks before and after impregnation or by burning the wax out of the band at a minimum temperature of 1100°F for 30 minutes in a protective atmosphere. In the event of a dispute concerning the wax content of any band lot, the wax content shall be checked by a method approved by the contracting officer.

4.6 Rejection.

4.6.1 Examination defects.- Any sample unit having one or more defects shall be rejected. If the number of nonconforming sample units exceeds the acceptance number specified in paragraph 4.3.5 for the sample size, the entire lot shall be rejected.

4.6.2 Test failures.- A lot shall be rejected for failure to meet any of the test requirements when tested in accordance with 4.5.

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5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.- Cleaning, drying, preservation and packaging shall be such that adequate protection is provided against corrosion, deterioration and damage during direct shipment from the supply source to the first receiving activity for immediate use or for controlled humidity storage.

5.2 Packing.- The packaged blanks shall be packed by sizes in boxes or cartons or on pallets in such a manner as to permit acceptance by carrier for safe transportation at the lowest rate and to afford maximum protection from normal hazards of transportation.

5.3 Marking.- Each box, carton or pallet shall be marked with the following information: quantity of blanks, size of blank, lot number, the contract or order number and such additional designations as may be specified in the contract or order.

6. NOTES

6.1 Intended use.- The rotating band blanks are intended primarily for use on artillery projectiles.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Size of the projectile for which intended.
(c) Required dimensions.
(d) Required tolerances.

Custodian:

Army - MU

Preparing activity:

Army - MU

Project No. 1395-A025

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